Interim Report
Abductive Event Reasoning: Towards Real-World Event Causal Inference for Large Language Models
SemEval-2026 Task 12

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Abstract

This paper presents an interim report on our ongoing work for SemEval 2026 Task 12: Abductive Event Reasoning: Towards Real-World Event Causal Inference for Large Language Models. The task focuses on identifying the most plausible causal explanation for a given real-world event based on contextual documents. We describe the task formulation, datasets, and baseline systems, including both large language models and encoder-based approaches. We further examine different prompting strategies and report preliminary experimental results, highlighting the strengths and limitations of current models. Finally, we outline planned directions for improving abductive reasoning performance in future work.

1 Introduction

Abductive reasoning requires models to infer the most plausible explanation for a given incomplete context. SemEval-2026 Task 12 focuses on multiple-choice abductive reasoning across narrative scenarios and noisy distractor documents. This report presents our methods, baseline experimentation, and preliminary findings.

2 Task Description

SemEval 2026 Task 12 focuses on the problem of Abductive Event Reasoning, where the goal is to identify the most plausible cause of a given real-world event. For each instance, the task provides a target event description along with a set of retrieved contextual documents, which may include both relevant evidence and distractor information.

Based on the provided context, systems are required to answer a multiple-choice question by selecting the correct causal explanation from four candidate options. The candidate explanations are designed to be semantically plausible, increasing the difficulty of the task and requiring models to perform fine-grained causal reasoning rather than surface-level pattern matching.

System performance is evaluated using accuracy, defined as the proportion of instances for which the model correctly identifies the ground-truth causal explanation.

3 Dataset

Our experiments are conducted on the datasets released for SemEval 2026 Task 12.

Validation Set (val_data). The official validation set is used for systematic evaluation and comparison of baseline models.

Sample Set (sample_data). The sample set is a smaller subset designed for rapid experimentation and prompt-level analysis.

Each instance consists of a target event description, a collection of retrieved contextual documents, and a multiple-choice question with four candidate causal hypotheses.

4 Baseline Models

We evaluate three representative baseline models: Qwen, LLaMA, and DeBERTa.

4.1 Qwen

We use *Qwen3-30B-A3B-Thinking-2507-GGUF* as our primary LLM baseline. Zero-shot English prompting yields the best results.

4.2 LLaMA

We employ *LLaMA-3.1-8B-Instruct* as a lightweight instruction-following baseline.

4.3 DeBERTa

As an encoder-based baseline, we reformulate the task as independent (question, option) pair scoring and select the highest-scoring option.

Model	Qwen	LLaMA	DeBERTa
Accuracy (%)	77.11	∼61.67	∼83.00

Table 1: Preliminary results on val_data.

Model	Qwen	LLaMA	DeBERTa
Accuracy (%)	71.25	∼60.00	∼60.00

Table 2: Results on sample_data.

5 Prompting Strategies

We explore zero-shot, few-shot, and multi-expert prompting. We additionally compare English and Chinese prompt variants to analyze cross-lingual sensitivity.

6 Preliminary Results

6.1 Validation Dataset

6.2 Sample Dataset

6.3 Qwen Prompting Performance

7 Discussion

Qwen achieves the strongest results, particularly under zero-shot English prompting. LLaMA exhibits higher output variability, while DeBERTa captures surface-level relevance but struggles with implicit causal reasoning.

8 Next Steps

We plan to extend our system with structured and multi-expert reasoning:

- Hybrid LLM–encoder systems
- RLHF-style feedback for explanation ranking
- Retrieval-enhanced multi-expert ensembles

9 Conclusion

This interim report highlights the challenges of abductive event reasoning under noisy evidence conditions. Our results suggest that large instruction-tuned models provide strong baselines, with further gains expected from hybrid and structured reasoning approaches.

A Chain-of-Thought Methods

B Prompt Examples

Prompting Strategy	Accuracy (%)
Zero-shot, English	71.25
Zero-shot, Chinese	66.25
Few-shot, English	62.29
Multi-Expert, English	60.00

Table 3: Qwen performance across prompting strategies on sample_data.

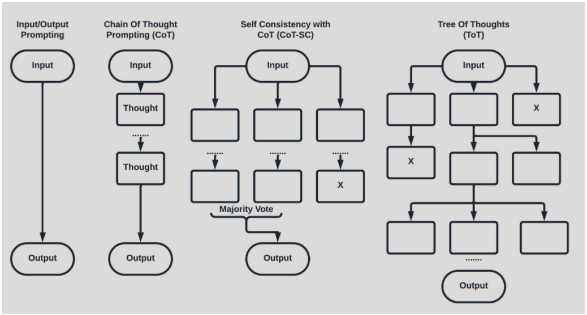


Figure 1: Planned multi-expert chain-of-thought strategy.

```
        self.expert_prompts = {
            "Timeline & Reference Specialist": (
                f"You are the Timeline & Reference Specialist. {distractor_warning}\n"
                "Your goal is to prevent chronological errors and resolve linguistic references.\n"
                "1. **Timeline Construction:** List the exact sequence of events mentioned in the text. "
                "Determine if the 'Option' occurred BEFORE or AFTER the 'Target Event'. "
                "If an option describes an event that happened *after* the target event, it cannot be the cause.\n"
                "2. **Reference Resolution:** Explicitly resolve pronouns and temporal markers. "
                "If the text says 'Since then' or 'Following this', clearly state WHICH specific event it refers to based on the paragraph structure.\n"
                "3. **Verdict:** Discard any option that violates the timeline or reference logic."
            ),
            "Critical Skeptic": (
                f"You are the Critical Skeptic. {distractor_warning}\n"
                "Your role is to champion the 'None of the above' or 'Other' possibility.\n"
                "1. **Attack the Options:** For each option (A, B, C, D), rigorously test if the text *explicitly* supports it as the CAUSE.\n"
                "2. **Logical Gap Detection:** If the text says 'Event X happened, and *then* Event Y happened', do not assume X caused Y unless the text explicitly links them (Post Hoc Fallacy).\n"
                "3. **Defense of the Null Hypothesis:** If options A, C, and D have flaws, YOU must argue why 'Option B' (or 'None of the others') is the most accurate choice. "
                "Do NOT force a selection if the evidence is weak."
            ),
            "Causal Alignment Specialist": (
                f"You are the Causal Alignment Specialist. {distractor_warning}\n"
                "Your goal is to distinguish between 'Root Causes' (Context) and 'Proximate Triggers' (Direct Causes).\n"
                "1. **Proximate vs. Root:** \n"
                "  - Example: 'Deforestation caused the fires' (Root Cause) vs 'International pressure caused the President to ban fires' (Proximate Trigger).\n"
                "  - If the question asks about the President's action, look for the *political* trigger, not the *scientific* cause of the problem.\n"
                "2. **Attribution Check:** Does the text attribute the event to a specific actor or report? "
                "If the text says 'The President responded to X', but the option says 'The President responded to Y', reject the option."
            )
        }
```

Figure 2: Multi-expert prompt used in experiments.